Week 51: Honest measurement ranges

Adult guide built from the problem files · Unpiloted. The strip kit is physically unrehearsed.

This view repeats, by problem, the answer key of the print guide, week-51-facilitator.pdf, whose source is editable/facilitator-src/content.json. Change both together; check_guide_copy.py compares them.

Mathematical overview

Guaranteed intervals. If $a \in [L, U]$ and $b \in [V, W]$, then $a + b \in [L + V, U + W]$ and $a - b \in [L - W, U - V]$. When a and b may vary freely in their separate closed intervals, these bounds are tight and both endpoints can occur. This is deterministic uncertainty: no probability distribution or confidence percentage is assumed.

“Always reaches r” means the minimum is at least r; “always fits within s” means the maximum is at most s. Equality is allowed in both. Except when a problem explicitly asks for whole-unit pairs, hidden endpoints may lie anywhere in the band, including between ticks and at the edges. The cards stipulate guaranteed bounds; the ruler itself does not establish an instrument error bound.

Dependencies matter. Repeated A is one unchanged quantity. Thus (A+4)-(A+1)=3 and (A+B)-A=B. Using separate ranges for repeated copies can enclose the answer too broadly. A known relation such as A+B=12 restricts possible endpoint combinations. The elementary formulas remain enclosures, but need not remain tight when the variables are linked.

Discovery by band. K-1 builds extreme examples and tries to defeat “always” claims. Grades 2-3 designs the widest safe card and compares shared versus separate strips. Grades 4-5 tightens bounds using an exact total and shared-piece cancellation. Testing examples can disprove a claim; endpoint arguments prove a guarantee and its sharpness.

Prepare and launch

Per pair or trio: printed pages, pencils and one strip set from **week-51-materials.pdf** on card stock at actual size (two sheets for a table of two pairs). **One unit is 10 mm**; each main 0-16 ruler and each sheet row is 160 mm long. A set has two gray strips of each whole length from 1 to 15 units and two white add-ons each of 4 and 1 units for the shared-A page; every extreme in the keys is a whole length. To take a length from a band, lay a strip along it from 0 and check that the end lands inside; join strips end to end on a 0-16 ruler. Repeated A means two equal strips, or one strip used twice. Cuisenaire rods also fit. Cutting takes about 10 minutes per set, less with a paper trimmer; the sets are reusable. The kit is physically unrehearsed.

Readiness. K-1 needs matching endpoints and counting small lengths, with adult reading. Older children need addition/subtraction and comparison of intervals. Negative numbers are unnecessary: in every printed gap task, the designated upper/longer join stays to the right. Keep an adult at each of the three tables.

Launch, 4 minutes. Let children find the strips whose ends land inside a band. Show A from 2 to 3 and B from 1 to 2 using the printed visual: A=2 and B=2, joined gap-free from 0, end at 4, inside the A + B band. Ask for a shorter join and a longer one; the shortest is 3 and the longest 5. Ask, “Can you choose allowed lengths that break my promise?”

Flexible hour. 0-5 launch; 5-22 joins and adversarial tests; 22-25 stand and form shortest/longest human strips; 25-43 choose cards or compare shared pieces; 43-55 exchange guarantees with witnesses; 55-60 reset. Keep uncertain ranges visibly distinct from one actual trial.

Problems and routes

Problem	K–1	2–3	4–5	route
Shortest and longest join	P1	P1	P1	first
Which pairs keep the promise	P2	P2	P2	first
A card for B that keeps the join in 7 to 9	P3	P3	P3	first
Which strip to uncover		P4	P4	first
Biggest and smallest gap between two strips	P4	P5	P5	return
A gap of 3 two ways	P5			return
The middle values do not give one exact gap		P6	P6	return
The same A twice, or A against a separate B	P6	P7	P7	return
Every whole-unit pair that fills 9	P7	P8		return
A known total narrows the gap			P8	return
Remove the very same A			P9	return

First route. K–1 P1–P3; Grades 2–3 P1–P4; Grades 4–5 P1–P4. **Return visits.** K–1 P4–P7; Grades 2–3 P5–P8; Grades 4–5 P5–P9.

Key, one entry per problem

Shortest and longest join

K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1 · first route · wording differs by band

Idea. A join of A in [4,6] and B in [3,4] can end anywhere in [7,10] and nowhere else: the extremes pair shortest with shortest and longest with longest.

A is 4-6 and B is 3-4, so the tight join range is **[7,10]**. Extremes are 4+3=7 and 6+4=10. K-1 can make exactly 8 with 4+4 or 5+3. In the older task, either extreme defeats the claimed 8-9 guarantee. Every sum lies in [7,10], and both endpoints are allowed, so neither bound can improve.

Which pairs keep the promise

K–1 P2 · Grades 2–3 P2 · Grades 4–5 P2 · first route · wording differs by band

Idea. A promise about a join holds exactly when both extreme joins keep it; one extreme setting defeats it otherwise.

Use only the three printed row pairs, not arbitrary cross-pairings.

Printed pair	Tight sum	Always reaches 8	Also fits within 10
A 4-6 with B 3-4	[7,10]	No: 4+3=7	Yes
C 4-5 with D 4-5	[8,10]	Yes	Yes
E 6-8 with F 2-4	[8,12]	Yes	No: 8+4=12

K-1 asks only about reaching 8, so **C with D and E with F** both work. Grades 2-3 and 4-5 require both promises, so **C with D alone** works. Reaching exactly 8 and ending exactly 10 count as successes. The endpoint witnesses in the table defeat the rejected guarantees.

A card for B that keeps the join in 7 to 9

K–1 P3 · Grades 2–3 P3 · Grades 4–5 P3 · first route · wording differs by band

Idea. With A in [3,4], B's card must start at least 4 (A might be 3) and end at most 5 (A might be 4), so [4,5] is the unique widest safe card.

Choose **B from 4 to 5**. With A from 3 to 4, all sums then lie in [7,9]. K-1's 3-4 card fails at 3+3=6; its 5-6 card fails at 4+6=10. In the older task, B's lower endpoint must be at least 4, because A might be 3; B's upper endpoint must be at most 5, because A might be 4. Hence [4,5] is the unique widest interval.

Which strip to uncover

Grades 2–3 P4 · Grades 4–5 P4 · first route

Idea. Learning one length exactly leaves the other strip's width of uncertainty, so uncover the strip with the wider range.

Uncover **A**. Knowing A=a leaves [a+3,a+4], width 1. Knowing B=b leaves [b+4,b+6], width 2. Revealing the strip with the wider original range removes more uncertainty; the exact revealed value shifts the remaining range but does not change its width.

Biggest and smallest gap between two strips

K–1 P4 · Grades 2–3 P5 · Grades 4–5 P5 · return route · wording differs by band

Idea. The gap long minus short is smallest at the shortest long strip and the longest short strip, and largest the other way round.

K–1: The long strip is 7-9 and the short strip 5-6. Largest gap **4** occurs at 9-5; smallest gap **1** occurs at 7-6. Each end moves separately. Subtracting shortest from shortest would not give the smallest gap.

Grades 2–3, Grades 4–5: Long strip [13,15] minus short strip [10,12] has tight gap **[1,5]**. Use 13-12=1 and 15-10=5 to show both bounds are attained. Every allowed long length is at least 13 and every short one at most 12, so the difference is positive.

A gap of 3 two ways

K–1 P5 · return route

Idea. Different settings can give the same gap: 8-5 and 9-6 both make 3.

Two whole-unit ways to make gap 3 are **8-5 and 9-6**. With continuous endpoints, there are also intermediate choices such as 8.5-5.5; the prompt does not restrict this problem to whole units.

The middle values do not give one exact gap

Grades 2–3 P6 · Grades 4–5 P6 · return route

Idea. 14-11=3 is one possible gap, not a guaranteed one: the settings at the two bounds give 1 and 5.

Subtracting middle values gives 14-11=3, which is possible but not guaranteed. Either 13 and 12 (gap 1) or 15 and 10 (gap 5) defeats the assertion of one exact gap. No probabilistic claim makes the midpoint more authoritative.

The same A twice, or A against a separate B

K–1 P6 · Grades 2–3 P7 · Grades 4–5 P7 · return route · wording differs by band

Idea. (A+4)-(A+1) is exactly 3 because the same A cancels, while (A+4)-(B+1) with A and B separately in [4,6] spreads over [1,5].

Top comparison: A+4 versus that same A+1. The end gap is always **3**; both smallest and largest gaps are 3, because the shared A cancels.

Bottom comparison: A+4 versus B+1, with A and B each separately allowed from 4 to 6. The tight gap is **[1,5]**. Minimum: A=4,B=6 gives 8-7=1. Maximum: A=6,B=4 gives 10-5=5. The two extremes use different values of A, since

each try may use new lengths. Intermediate values are possible too. The diagram shows one setting, not a promise that the shaded pieces always retain their drawn lengths.

Every whole-unit pair that fills 9

K–1 P7 · Grades 2–3 P8 · return route

Idea. An exact total links the strips: B = 9 - A, so each whole A gives one pair, kept only if B fits its card; A = 3 and A = 7 fail.

With A from 3 to 7, B from 3 to 5, whole-unit lengths, and total 9, the complete list of ordered pairs (A,B) is **(4,5), (5,4), (6,3)**. A=3 would need B=6 and A=7 would need B=2, both off B's card; for A=4, 5 and 6, B=9-A fits its card. Four boards are printed for three pairs, so the child decides when the list is complete. The exact-sum condition links the strips.

A known total narrows the gap

Grades 4–5 P8 · return route

Idea. A+B=12 with B in [3,7] forces A into [5,9], so A is in [7,9] and the gap 2A-12 runs from 2 to 6; A=10 is discarded.

$A \in [7, 10]$, $B \in [3, 7]$ and A+B=12 imply B=12-A. The B card restricts A to [5,9]; intersecting with A's card gives $A \in [7, 9]$ and consequently $B \in [3, 5]$. Since A is always larger, the end gap is A-B=2A-12, ranging from **2 to 6**. Witnesses are (A,B)=(7,5) and (9,3). The discarded choice A=10 would require B=2 and is not allowed.

Remove the very same A

Grades 4–5 P9 · return route

Idea. (A+B)-A is exactly B, in [3,4]; subtracting the ranges [7,10] and [4,6] separately gives the valid but loose [1,6].

The tight remainder range is **[3,4]**, exactly B's original range. Physically remove the same A that was joined: all that remains is B. The claimed [1,6] is a valid but unnecessarily wide enclosure obtained by treating the total and A as freely variable. Its extreme combinations cannot occur together: a total of 7 requires A=4,B=3, while a total of 10 requires A=6,B=4.

Optional hints and facilitation checks

For a guarantee, ask the partner to choose the most awkward permitted ends. For a tight bound, ask for a setting that actually reaches it. For dependency, keep a finger or matching label on the same A. If the group is still comparing physical extremes, stop there rather than require interval notation or algebra.

Source and verification

NIST Digital Library of Mathematical Functions, **§3.1(ii)**, "Interval Arithmetic," equation (3.1.6): official reference (dlmf.nist.gov/3.1.ii). This is the current interval-arithmetic section; §3.1(v) is error measures. The strip endpoint calculations, repeated-variable cancellations and constrained-total cases here are independently derived. All dimensions were digitally checked; strip handling and the hour plan are unpiloted proposals.